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sEMG-based continuous estimation of grasp movements by long-short term memory network

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ABSTRACT

Controlling a robotic hand or arm by surface Electromyographic signals (sEMG) is an important research direction. The present pattern-recognition-based control strategy can realize some myoelectric control but it is not as smooth as human hand. In this paper, we proposed a continuous estimation method for 6 daily grasp movements by Long-Short Term Memory Network (LSTM). In addition, we compared Sparse Gaussian Processes using Pseudo-inputs (SPGP) and Radial Basis Function Neural Network (RBF) with LSTM for continuous estimation of 6 grasp movements. These methods are tested on the NinaPro dataset and evaluated by Pearson Correlation Coefficient (CC) as well as Root Mean Square Error (RMSE) and Normalized Root Mean Square Error(NRMSE) between real and estimated joint angles. They can estimate 20 joint angles in hand synchronously. The average CC of LSTM (0.8402 ± 0.0009) is significantly higher than RBF (0.774 ± 0.0018, $p = 0.0124$) and SPGP (0.7883 ± 0.0005, $p = 0.0075$). The RMSE of LSTM ($(5.89 \pm 0.95)^\circ$) is less than RBF ($(8.08 \pm 2.911)^\circ$, $p = 0.0213$) and SPGP ($(7.16 \pm 1.178)^\circ$, $p = 0.0495$) for the 6 movements. In addition, the NRMSE of LSTM ($15.27\% \pm 0.02\%$) is less than RBF ($19.84\% \pm 0.06\%$, $p = 0.0033$) and SPGP ($18.11\% \pm 0.02\%$, $p = 0.0117$).

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1. Introduction

sEMG contains the combined effects of superficial muscle EMG and nerve trunk electrical activity on the surface of the skin, which can reflect the activity of the neuromuscular to some extent [1]. Therefore, it can be taken as a bridge to establish the relationship between machine and human motion [2]. The concept of EMG-based control used in human-machine interaction can be traced back to 1970s [3,4]. The present myoelectric control methods include pattern-recognition-based methods, musculoskeletal-model-based methods as well as NMF-based methods, etc. But they have some limitations. For example, the computation of musculoskeletal-model-based methods are very complex. Although the pattern-recognition-based myoelectric control can satisfy some practical requirement, it is not as smooth as real human hand [4–6]. Recently, continuous motion estimation by sEMG has been a popular research topic [7].

In the last two decades, static gestures classification has procured good results. Weidong Geng et al. (2016) proposed to use sEMG images to recognize gestures, and the accuracy can reach over 95% [8]. From ten years ago, many attempts have been taken to achieve continuous motion estimation in order to enable the control of robotic hands or arms naturally. Qichuan Qing et al. (2011) developed a Hill-based muscle model (HMM) to estimate continuous joint angles displacement and velocity of elbow flexion and extension [9]. Then in 2014, Jianda Han extended HMM to form a state-space HMM and used a closed-loop prediction–correction approach to estimate the continuous movements based on sEMG

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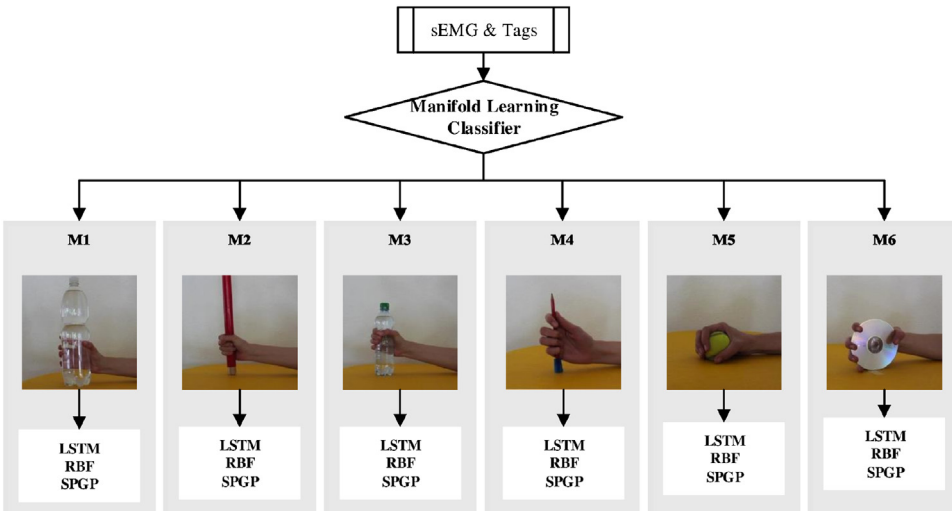


Fig. 1. M1: Large diameter grasp; M2: Small diameter grasp; M3: Medium diameter grasp; M4: Stick grasp; M5: Power sphere grasp; M6: Power disk grasp.

[10]. Some works also focused on the estimation of joint angles directly. For example, Yee Mon Aung (2012) proposed a back propagation neural network (BPNN) to estimate three shoulder joint angles and one elbow joint angle [11,12]. Hahne et al. (2014) compared the performance of four regression methods in simultaneous and proportional myoelectric control, which are linear regression, mixture of linear experts, multilayer-perceptron and kernel ridge regression [13]. Both Qin Zhang [14] and Aadeel Akhtar [15] realized the continuous estimation of shoulder and elbow or distal arm joints in 2017. But these works mainly achieved the motion estimation of the upper limb or wrist, which involves 3–4 degrees of freedom (DOF). Since 2013, more efforts have been taken on the continuous estimation of hand movements, involving more DOFs. For example, Jimson Ngeo et al. (2013) proposed an artificial neural network (ANN) and EMG-Muscle Activation Model to estimate the angle of index metacarpal, index proximal and index distal [16,17]. And then in 2014, they compared multi-output convolved Gaussian Process (Multi-output Full GP) with ANN for the continuous estimation of 16 joint angles in hand, including metacarpal joint, proximal joint and distal joint of each finger [18,19]. But the CC had an obvious difference in different joints, and the average value was less than 0.8. After then, Agamemnon Krasoulis attempted to estimate 22 joint angles in hand together, and the R^2 within-movement generalization was 0.9 while across-movement generalization was less 0.6 in average [20]. Apart from these, Luzheng Bi et al. (2019) reported a review of the sEMG-based continuous estimation of human upper limb motion [21]. The review comprised the models and algorithms for sEMG-based continuous estimation or kinematic parameters, as well as the performance measurement for system validation.

According to the present works, the features of sEMG is usually extracted before regression experinment, which can be divided into three kinds, containing time domain features (TD), frequency domain features (FD) and time-frequency domain features (TFD) [22–24]. Generally, TD is used widely in both classification and regression because of its smaller computational complexity than FD and TFD. Variance (VAR) [25–27], mean absolute value (MAV) [26,28,29] and root mean square (RMS) [11,30–33] etc. are also commonly used in the features extraction. And some works also used FD features like autoregressive coefficients (AR) [34]. In addition, a few works used the combination of TD features and FD features for classification or the estimation of hand movements [35–37]. However, at present, which feature or combination of features can result in the best accuracy is usually determined by a trial-and-error approach [21].

In this study, we estimate 6 grasp movements using three methods based on sEMG. Both sEMG and joint angles were derived from the public dataset, NinaPro. After comparing the performance of several features, we choose MAV as sEMG feature.

2. Methods

2.1. Database

NinaPro is a public dataset firstly released by Manfredo Atzori et al. in 2014 [38]. It contains 7 databases, with each database containing 3 exercises.[2] These databases include sEMG, hand kinematics and dynamics from intact subjects or amputees when doing hand motions with different devices. Each movement was repeated 6 times, with each repetition lasting 5 s and resting 3 s.

In this study, we design experiments on 6 daily grasp movements shown in Fig. 1 from 5 subjects (3 men and 2 women, between 25 and 30 years old, 4 right-handedness and 1 left-handedness) in DB2-Exercise C. These 6 movements are selected according to different shapes and different diameters of the objects. They are representative because the joints involved are different when grasping objects of different shapes or diameters,and the motion range of each joint is also different. The 5 subjects are chosen from young adults with different genders, different weights, and different handedness.

The 6 grasp movements are classified by a matured manifold learning classifier before the regression task as shown in Fig. 1. The input vectors of manifold learning classifier are 12 channels sEMG features (MAV, see formula (1)). The tags of the movements adopted here are determined in the NinaPro dataset.

In DB2, *CyberGlove II* measured 22 joint angles. The position and number of sensors are shown in Fig. 2 and the corresponding joints are listed in Table 1. But in this study, we only researched on grasp movements, thus No.1–No.20 joints were considered. sEMG signals with 12 channels were recorded by *Trigno wireless (Delsys)*. The 12 electrodes were stuck on the arm, as 1–8 electrodes equally spaced around the forearm, 9 and 10 spaced on the surface of flexor dig-

[2] Each exercise includes several movements. More details about the databases can be referred from http://ninapro.hevs.ch/.

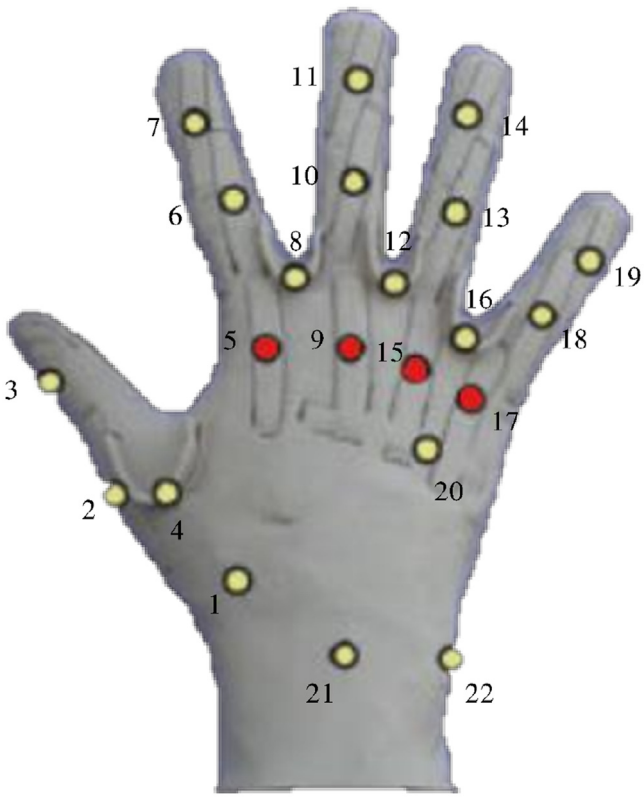


Fig. 2. The map and number of sensors in *CyberGlove II*.

Table 1
The Joints Corresponding to the Sensors in *CyberGlove II*

Number	Joint	Number	Joint
1	Thumb Roll	**12**	Middle-Ring Abduction
2	Thumb Metacarpal	**13**	Ring Metacarpal
3	Thumb Proximal	**14**	Ring Proximal
4	Thumb-Index Abduction	**15**	Ring Distal
5	Index Metacarpal	**16**	Ring-Pinky Abduction
6	Index Proximal	**17**	Pinky Metacarpal
7	Index Distal	**18**	Pinky Proximal
8	Index-Middle Abduction	**19**	Pinky Distal
9	Middle Metacarpal	**20**	Palm Flex
10	Middle Proximal	**21**	Wrist Pitch
11	Middle Distal	**22**	Wrist Yaw

itorum superficialis and digitorum superficial extenso, 11 and 12 measuring signals from Biceps Brachii and Triceps Brachii.

2.2. Preprocessing and features extraction

Due to the low sampling accuracy of the glove, the joint angle signals collected are stepped, but considering the smoothness of the human motion, these signals should be smoothed in the training procedure firstly. In order to avoid the phase delay between the angle signals before smoothed and after smoothed, 4th order 2 Hz zero-phase filter was used [39]. Then the band-pass Butterworth filter with the 2nd order 5 Hz–450 Hz was used to remove high-frequency noise in sEMG signals.

After filtering, MAV was used to extract sEMG features because of its uncomplicated compute and better performance. 200 ms sliding window, with 50% overlap, was used in features extraction here. The MAV of a segment k is calculated by adding the absolute value of all of the values and divided by the length of the segment as following [40,41]:

$$MAV_k = \frac{1}{N}\sum_{i=1}^{N}\left|x_i\right|, \tag{1}$$

where N is the length of the segment k, x_i is the values in k.

The sampling rate of *CyberGlove II* (20 Hz) is different from *Delsys* (2KHz), so sEMG should be synchronized with joint angles. Linear interpolation was used to make the sEMG features and joint angles correspond one-to-one in time and sampling size. Then according to the maximum and minimum of the features and angles, they were normalized to [−1, 1] in order to improve the training speed of networks and accelerate the convergence process of gradient descent.

2.3. Models

2.3.1. Long-short term memory (LSTM)

LSTM network firstly proposed by Hochreiter and Schmidhuber in 1997 [42] is a type of many-to-many Recurrent Neural Network (RNN) built to model temporal sequences and avoid long-term dependence problems. As one of the most widely used networks in time-series signals processing, LSTM has achieved good results in speech recognition, video analysis, and other fields. The continuous estimation based on sEMG signals is similar to the time-series signals. If the sEMG data at time t is regarded as a graph, the sequence data of $[t_0, t_1 .. t_n]$ can be approximated as video stream data.

RNN-based networks have a special chain structure, and the nodes in the chain structure are called memory cells. In a standard RNN, a memory cell is implemented by a simple neural network, but in the LSTM network, the implementation of the memory cell is more complex. In this study, the network was with 4 interactive layers in each cell [43]. The structure is shown as Fig. 3.

There are three gate units in a cell, called "forget gate layer", "input gate layer" and "output gate layer", respectively. These gates protect and control the state of a cell, then complete computing in four steps shown as (1)–(4) in Fig. 4.

Firstly, "forget gate layer" decides to discard what information with part (1) in Fig. 4. The function is:

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right). \tag{2}$$

Then part (2) in Fig. 4 of the cell identifies the information needed to be updated:

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right), \tag{3}$$

$$\tilde{C}_t = \tanh\left(W_C \cdot [h_{t-1}, x_t] + b_C\right). \tag{4}$$

Part (3) in Fig. 4 complete the update of cell state:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t. \tag{5}$$

At last, the output is computed based on cell state with part (4) in Fig. 4:

$$o_t = \sigma(W_o[f_t, x_t] + b_o), \tag{6}$$

$$h_t = o_t * \tanh(C_t), \tag{7}$$

where σ and *tanh* are functions, and W_i, W_C, W_o are the input layer, hidden layer and output layer weight matrices, respectively. b_i, b_C, b_o are the bias matrices of those layers, and i_i, f_t, o_t indicate vectors of three gates layers. C_t is the cell state vector.

In this study, the LSTM is set to three layers with every layer containing 128 hidden units. The input of this network is 12 channels sEMG features and the afterwards layer inherits the state of previous layer. At last, the output is 20 joint angles. The RMSE between the predicted and real angle is chosen as loss function while Adam

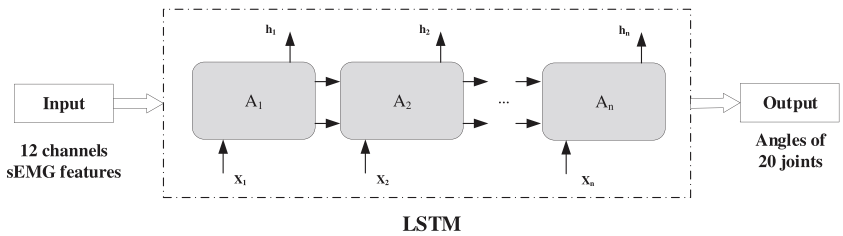


Fig. 3. The chain structure with repetitive modules of LSTM, where $A_1, A_2, \ldots, A_n$ indicate the modules of chain structure.

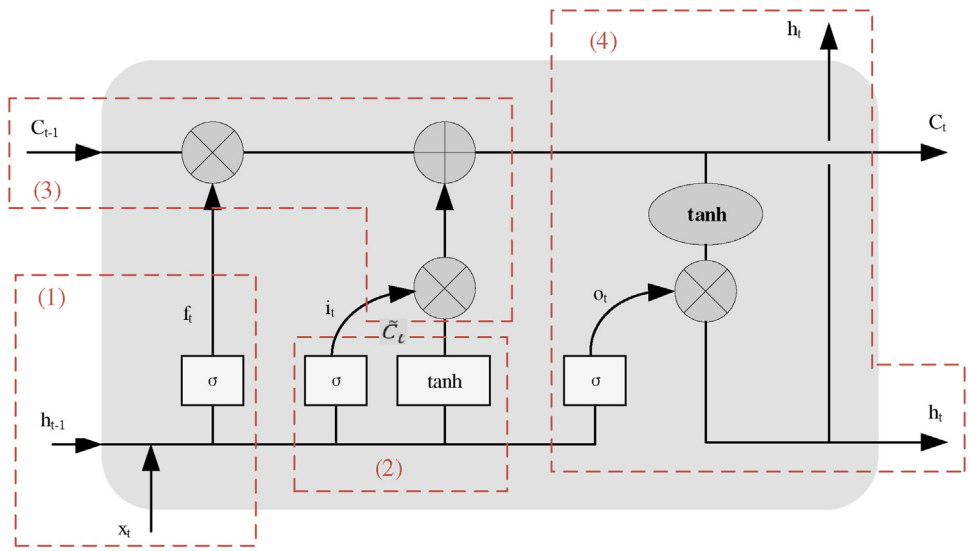


Fig. 4. The detailed structure in one module of LSTM.

optimizer iss considered. In order to prevent overfitting effectively, dropout is set to 0.3. Finally, a fully connected layer with an input of 128 and an output of 20 maps network internal parameters to 20 joint angles.

2.3.2. Radial basis function neural network (RBF)

RBF Neural Network belongs to the forward neural network and consists of an input layer, a hidden layer and an output layer [44]. The basic network structure is shown in Fig. 5.

The activation function used in this RBF Network is Gaussian function, which can be defined as following:

$$R\left(x_p - c_i\right) = \exp\left(-\frac{1}{2\sigma^2}\left\|x_p - c_i\right\|^2\right), \tag{8}$$

where $\left\|x_p - c_i\right\|$ is Euclidean norm; c_i is the center of Gaussian function; σ is variance of Gaussian function.

The output of network shown in Fig. 5 can be described as following:

$$y_i = \sum_{i=1}^{h} \omega_{ij} \exp\left(-\frac{1}{2\sigma^2}\left\|x_p - c_i\right\|^2\right), j = 1, 2, \cdots, n, \tag{9}$$

where $x_p = (x_1^p, x_2^p, \cdots, x_3^p)^T$ is the *pth* one in samples; $p = 1, 2, 3, \cdots, P$, P is the total number of samples; c_i is the center of the network hidden layer node; ω_{ij} is the weight from hidden layer to output layer; $i = 1, 2, 3, \cdots, h$, h is the number of hidden layer node; y_i is the actual output of the *ith* output node corresponding to input sample; σ is the variance of basis function, computed as follows:

$$\sigma = \frac{1}{P}\sum_{j}^{m}\left\|d_i - y_i c_i\right\|^2. \tag{10}$$

RBF network has the advantages of simple structure, fast convergence of training and learning, as well as approximating any nonlinear function.

In this study, the main function used is *newrb*() from MATLAB, which can design an approximate radial basis network. The 6 parameters of this function including *input vectors*, *target vectors*, *goal*, *spread*, *MN andDF* are detailed as follows.

- *input vectors* are 12 channels features;
- *taget vectors* are 20 joint angles, of which column number is the same to *input vectors*;
- *goal* is mean squared error goal with 0.0;
- *spread* is spread of radial basis functions with 1.0;
- *MN* is maximum number of neurons,which is the column number of *input vectors* or *taget vectors*;
- *DF* is number of neurons to add between displays with 25.

2.3.3. Sparse Gaussian processes using pseudo-inputs (SPGP)

SPGP is presented by Edward Snelson et al. in 2006 [45]. When a data set $D = \left\{\mathbf{X}, \mathbf{y}\right\}$ is given, where $\mathbf{X} = \{x_n\}_{n=1}^N$ is the input vector and $\mathbf{y} = \{x_n\}_{n=1}^N$ is the actual output. The function $y = f(x)$ is expected to be acquired. According to Gaussian process, supposing $\mathbf{y}$ obeys a normal distribution with a mean of 0, and the covariance K calculated by a kernel function, which we used here is RBF kernel function. By adjusting the observed data and hyperparameters, prediction can be made when considering a new input point x. The distribution of the target value at the point x is:

$$p\left(y|x, D, \theta\right) = N\left(y|k_x^T\left(K_N + \sigma^2 I\right)^{-1} y, K_{xx} - k_x^T\left(K_N + \sigma^2 I\right)^{-1} k_x + \sigma^2\right), \tag{11}$$

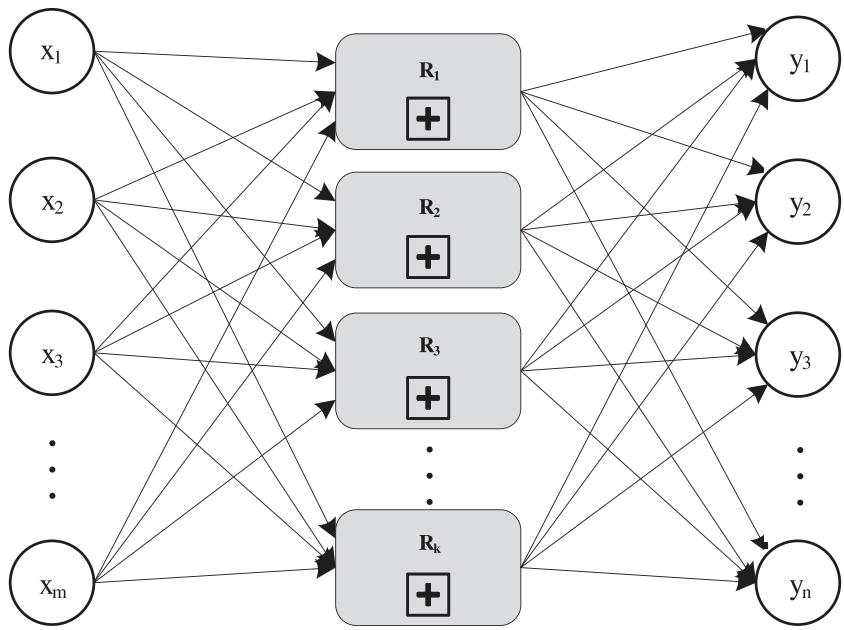


Fig. 5. The Basic Structure of RBF Network.

where $N(f|m, V)$ is a Gaussian distribution with mean m and covariance V; K_n is kernel function, $k_x = K(x_n, x)$ and $K_{xx} = K(x, x)$; θ and σ^2 is hyperparameters, $\sigma^2 I$ is noise. However, when the data sets are large, the computation of standard Gaussian is large as well as the calculation time. If the number of training samples is N, the dimension of the covariance matrix K is N^2, and the complexity of computational will be $\mathcal{O}\left(N^3\right)$ during the training. Therefore, the complexity of prediction computation will reach $\mathcal{O}\left(N^2\right)$ at least.

In SPGP, the entire training set N is replaced by a small number of input points M, where $M \ll N$. The pseudo-input is $\overline{X} = \left\{\bar{x}_m\right\}_{m=1}^{M}$, and corresponding pseudo target is $\bar{\mathbf{f}} = \left\{\bar{f}_m\right\}_{m=1}^{M}$, which is an ideal data set because there is no error from observation process. Thus the complexity of computation in training process will reduce to $\mathcal{O}\left(M^2 N\right)$, while that in predicting process will be $\mathcal{O}\left(M^2\right)$. This not only reduces the amount of calculation greatly, but also isn't affected by the number of samples in the training set during the prediction process.

In this algorithm for continuous estimation, We borrow the code written by the Sheffield Machine Learning Group for SPGP [45]. The input and target vectors are 12 channels sEMG features and 20 joint angles respectively, and the pseudo-input is 30 vectors from the entire training set. That is $M = 30$ here.

2.4. Cross-validation

Three-fold cross-validation is used to evaluate the model in this study [46]. The six repetitions of one movement are divided into three parts according to the timing. In the three-fold cross-validation, each of the three parts is used as the test set and a model is trained by using the remaining two parts. This validation can keep training and test set not only disjoint but as independent as possible [13]. Pearson Correlation Coefficient (CC), Root Mean Square Error (RMSE) and Normalized Root Mean Square Error (NRMSE) are used as the evaluation criterions. CC can measure the similarity between signal shapes, while RMSE can be used to evaluate the numerical error of amplitude. Considering the distinction between joints motion range, for some joint angle with a larger range and RMSE, its NRMSE may be smaller, which still means that the performance is good. So NRMSE is used to standardized the RMSE as a supplement. The calculation formulas are shown as follows:

$$CC = \frac{\sum_{i=1}^{N}\left(\theta_{est} - \bar{\theta}_{est}\right)\left(\theta_{real} - \bar{\theta}_{real}\right)}{\sqrt{\sum_{i=1}^{N}\left(\theta_{est} - \bar{\theta}_{est}\right)^2}\sqrt{\sum_{i=1}^{N}\left(\theta_{real} - \bar{\theta}_{real}\right)^2}}, \tag{12}$$

$$RMSE = \sqrt{\sum_{i}^{N}\frac{\left(\theta_{est} - \theta_{real}\right)^2}{N}}, \tag{13}$$

$$NRMSE = \frac{RMSE}{\theta_{\max} - \theta_{\min}}, \tag{14}$$

where θ_{est} and $\bar{\theta}_{est}$ are the estimated angle and their average, while θ_{real} and $\bar{\theta}_{real}$ are the real angle and their average. θ_{max} is the maximum of the real angle, and θ_{min} is the minimum of the real angle.

After cross-validation, three evaluation results will be obtained, and the average of three values will be taken.

3. Results

LSTM was built in Pytorch, while RBF and SPGP were built in Matlab to estimate 6 grasp movements of 5 subjects. The results of are shown in Fig. 6 and Figs. 11, 12, 13, 14, 15 in Appendix. Taking M1 (shown in Fig. 6) as an example, comparing the evaluation criteria from three methods, we conclude results as following.

- Based on the CC evaluation criterion, over 10 joint angles' CC values of LSTM can reach 0.9. Although in several joints, such as *No.* 11 and *No.* 19, the CC of SPGP (with 0.946 of *No.* 11 and 0.952 of *No.* 19) is higher than LSTM (with 0.943 of *No.* 11 and 0.947 of *No.* 19), the advantages are not obvious (with $\triangle CC < 0.01$[3]). However, in most joints, the CC of LSTM is significantly higher than RBF or SPGP. For example, in *No.* 18 or *No.* 20 joint, the CC of LSTM is higher than SPGP and RBF with more than 20%. Besides, as for *No.* 2, 3, 6, 9, 13 joint, the CC of LSTM is 5%–10% higher than SPGP or RBF. These indicate that the angles estimated by the LSTM match better with target angles than RBF and SPGP.

[3] The "△CC" represents the difference between CC of LSTM and CC of RBF or SPGP.

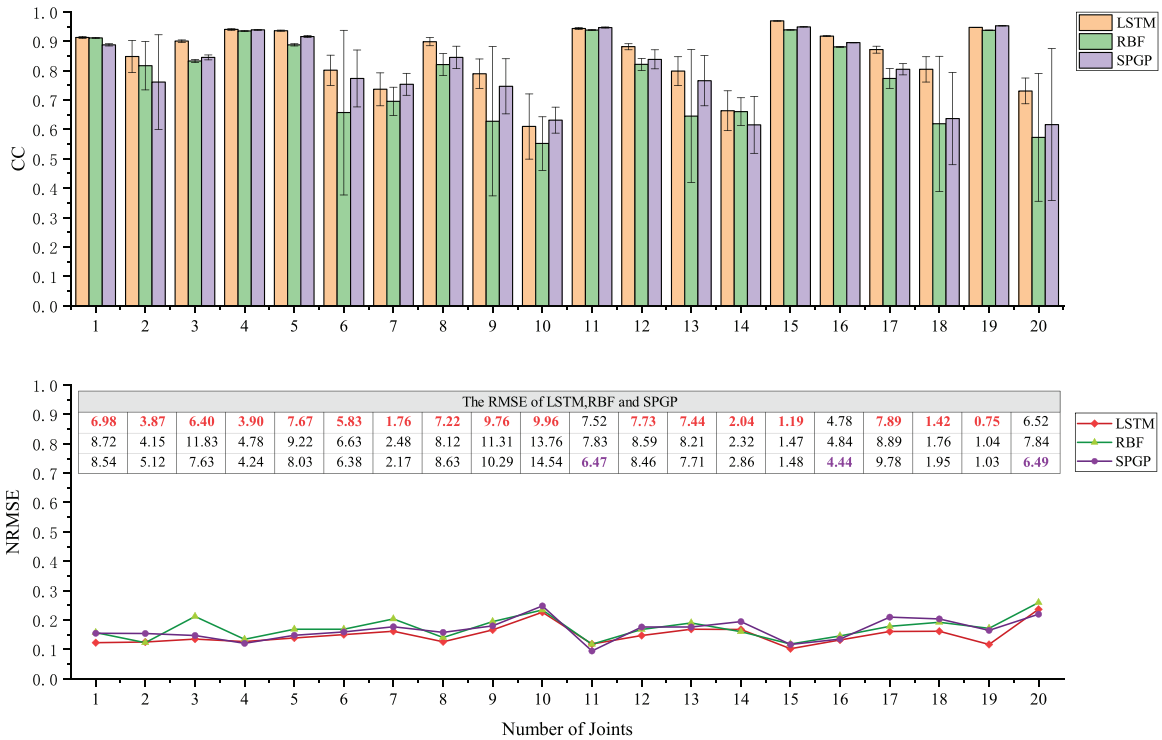


Fig. 6. The CC, RMSE and NRMSE of M1, the unit of "RMSE" is "◦".

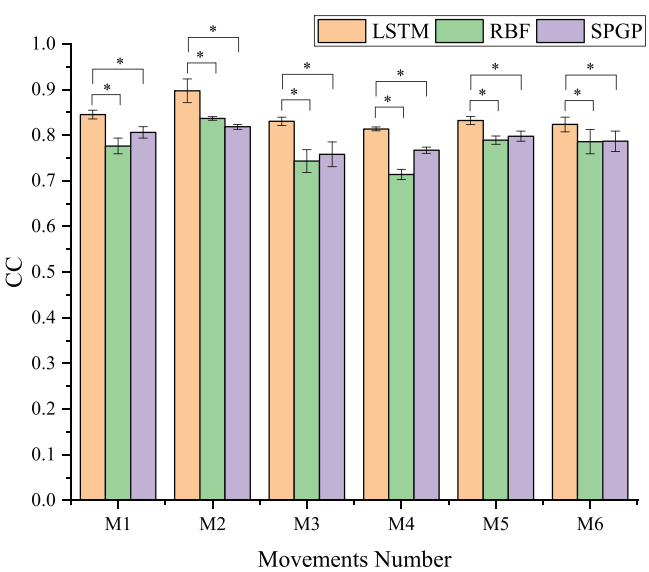


Fig. 7. Summary of the CC of LSTM, RBF and SPGP for 6 movements, where each CC is the average of the 20 joints. '*' indicates statistical significance ($p < 0.05$).

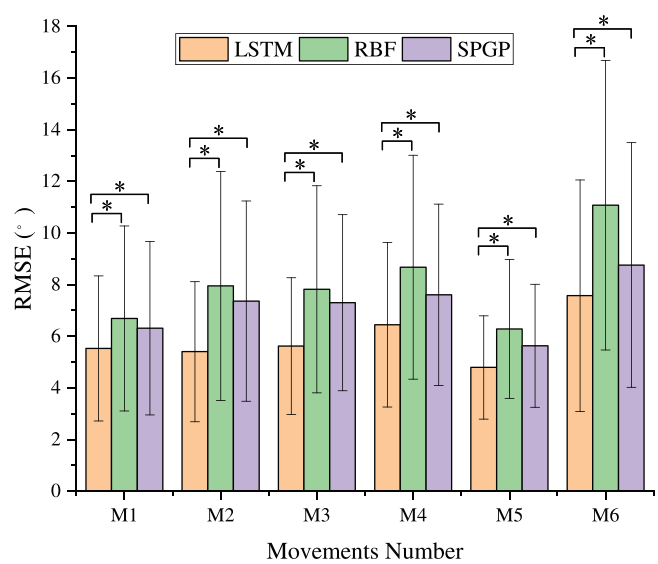


Fig. 8. Summary of the RMSE of LSTM, RBF and SPGP for 6 movements, where each RMSE is the average of the 20 joints. '*' indicates statistical significance ($p < 0.05$).

- The RMSE values of LSTM are in the range of 0.75°–9.96°, with an average of 5.531°. While the RMSE range of RBF and SPGP are respectively 1.04°–13.76° (average 6.689°) and 1.03°–14.54° (average 6.31°). In terms of the average, the RMSE of LSTM is 20% less than SPGP or RBF. Except for *No.* 11, 16, 20 joints, the RMSE values of LSTM are less than SPGP($\triangle$RMSE = (0.780 ± 1.487)°[4]) and RBF($\triangle$RMSE = (1.158 ± 1.271)°). This demonstrates that there are smaller differences in amplitude between real angles and estimated angles by LSTM.
- As a supplement, NRMSE can reflect the performance of different methods based on the range of real angles. The smaller the value of NRMSE, the better the methods perform. As shown in Fig. 6, for over three-quarters of the joints, the NRMSE values of LSTM (14.95% ± 3.3%) are less than SPGP (17.19% ± 3.68%) and RBF (16.68% ± 3.59%). It is verified that based on the motion range of each joint angle, the performance of LSTM for continuous estimation is better than RBF and SPGP. But for *No.* 2 and *No.* 14 joints, the NRMSE of RBF are less than LSTM with $\triangle$NRMSE ≈ 0.1%,[5] and

[4] The "$\triangle$RMSE" represents the difference between RMSE of LSTM and RMSE of RBF or SPGP.

[5] The "$\triangle$NRMSE" represents the difference between NRMSE of LSTM and NRMSE of RBF or SPGP.

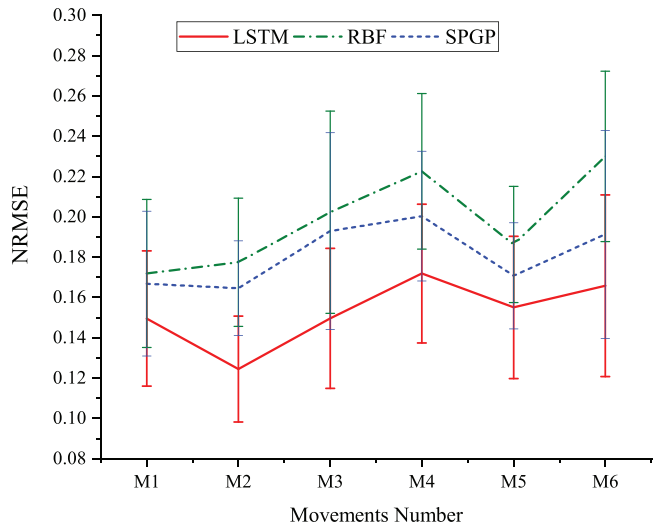


Fig. 9. Summary of the NRMSE of LSTM, RBF and SPGP for 6 movements.

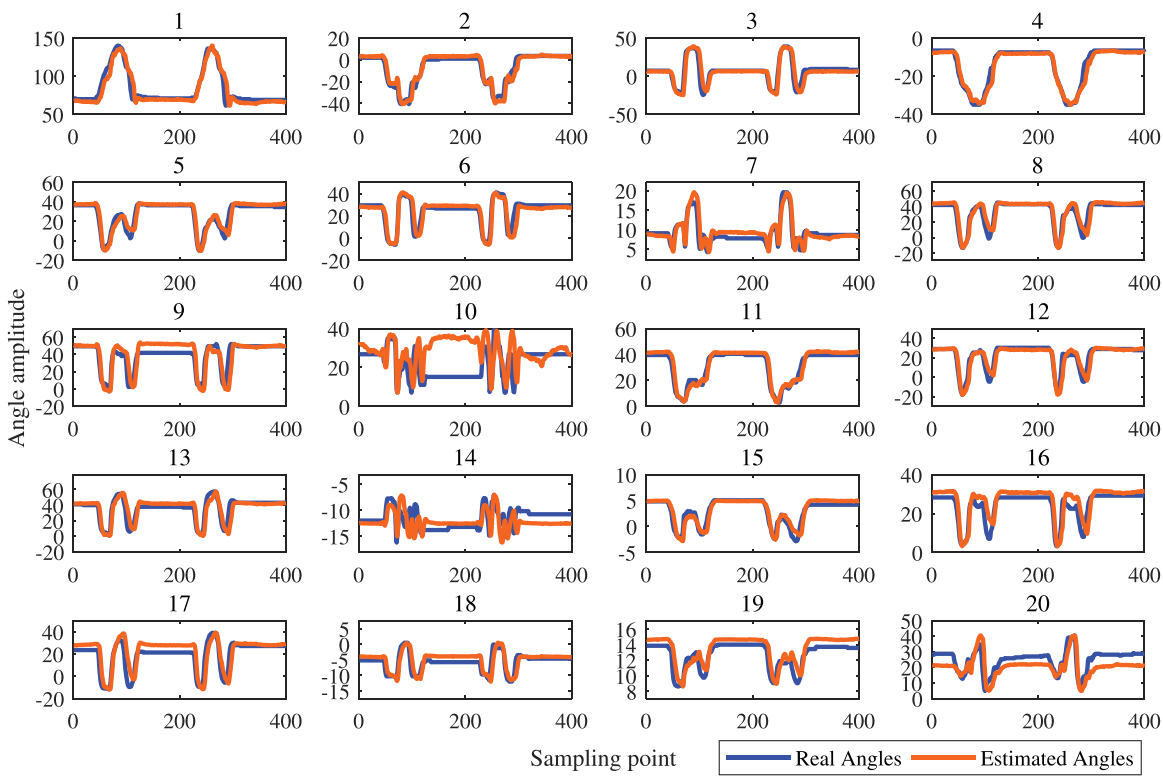


Fig. 10. Estimated and real joint angles for M1 of subject 1 (The unit of "Angle ampllitude" is degree measure, "◦").

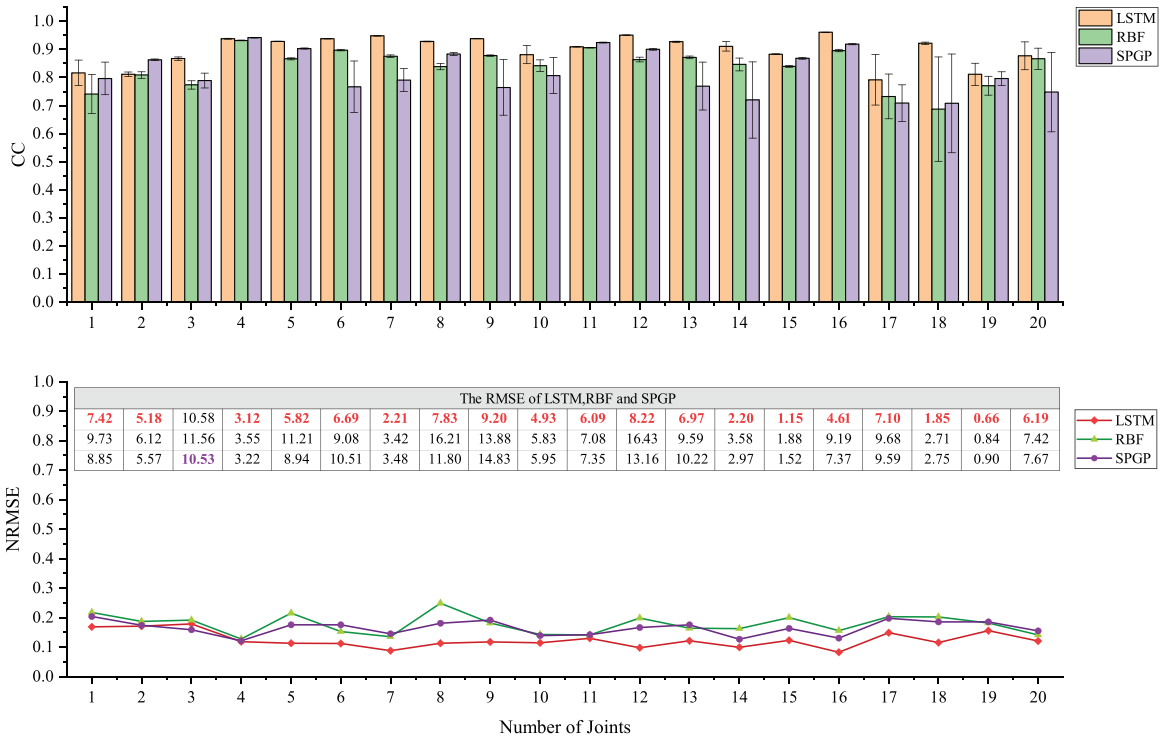
for *No*. 4, 11, 20 joints, the NRMSE of SPGP are less than LSTM with △NRMSE = 1.5% ± 0.7%.

At last, we make analysis of variance (ANOVA) for LSTM, RBF and SPGP. In ANOVA test, the CC with LSTM (0.8402 ± 0.0009) is higher than RBF (0.774 ± 0.0018, $p = 0.0124$) and SPGP (0.7883 ± 0.0005, $p = 0.0075$) (see Fig. 6). The RMSE with LSTM ((5.89 ± 0.95)◦) is less than RBF ((8.080 ± 2.911)◦, $p = 0.0213$) and SPGP ((7.160 ± 1.178)◦, $p = 0.0495$) (see Fig. 7). As shown in Fig. 10, the NRMSE of LSTM (15.27% ± 0.02%) is less than RBF (19.84% ± 0.06%, $p = 0.0033$) and SPGP (18.11% ± 0.02%, $p = 0.0117$).

Taking subject 1 as an example, Fig. 10 shows the linear correlation between the angles estimated by LSTM and the real measured angles for M1. And Figs. 16, 17, 18, 19, 20 in Appendix show the performance of LSTM for M2-M6.

4. Discussion

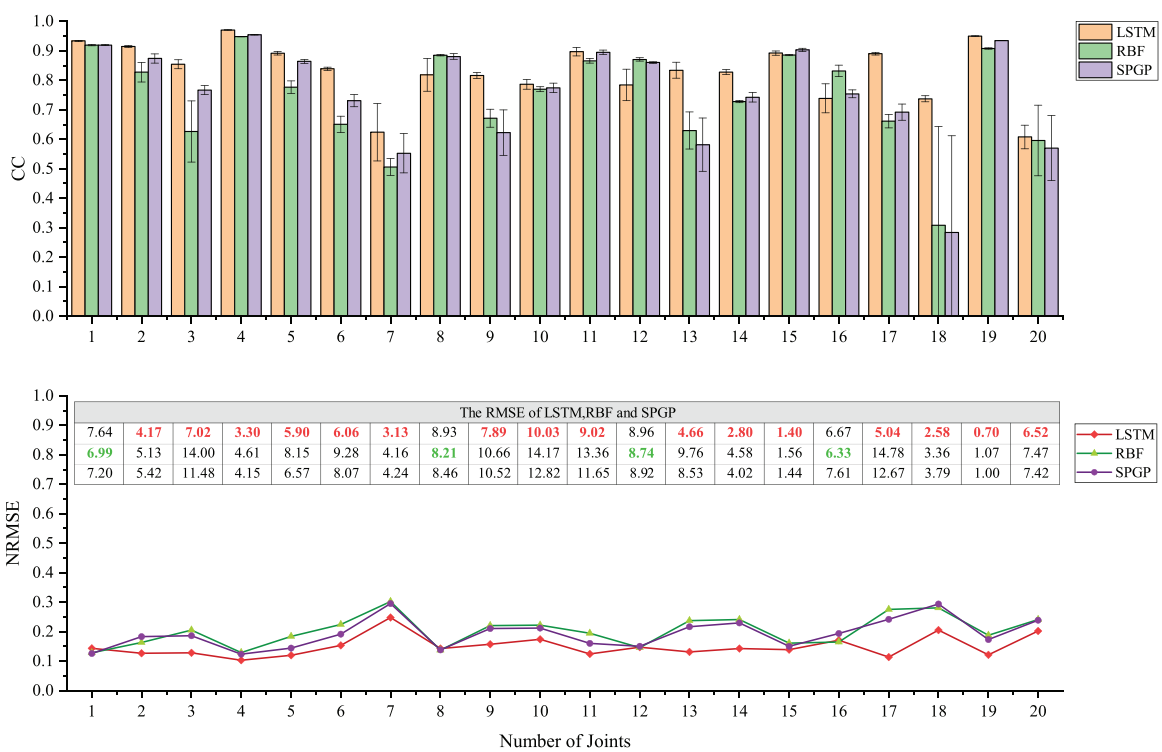
This study proposed a novel continuous estimation method based-on sEMG by LSTM. And we also compared the performance of two other algorithms for the continuous estimation of grasp movements. LSTM can perform better as well as faster in all 6 movements. Then we explored the difference between 6 movements with the same algorithm.



The RMSE of LSTM,RBF and SPGP

7.42	5.18	10.58	3.12	5.82	6.69	2.21	7.83	9.20	4.93	6.09	8.22	6.97	2.20	1.15	4.61	7.10	1.85	0.66	6.19
9.73	6.12	11.56	3.55	11.21	9.08	3.42	16.21	13.88	5.83	7.08	16.43	9.59	3.58	1.88	9.19	9.68	2.71	0.84	7.42
8.85	5.57	10.53	3.22	8.94	10.51	3.48	11.80	14.83	5.95	7.35	13.16	10.22	2.97	1.52	7.37	9.59	2.75	0.90	7.67

Fig. 11. The CC, RMSE and NRMSE of M2.



The RMSE of LSTM,RBF and SPGP

7.64	4.17	7.02	3.30	5.90	6.06	3.13	8.93	7.89	10.03	9.02	8.96	4.66	2.80	1.40	6.67	5.04	2.58	0.70	6.52
6.99	5.13	14.00	4.61	8.15	9.28	4.16	8.21	10.66	14.17	13.36	8.74	9.76	4.58	1.56	6.33	14.78	3.36	1.07	7.47
7.20	5.42	11.48	4.15	6.57	8.07	4.24	8.46	10.52	12.82	11.65	8.92	8.53	4.02	1.44	7.61	12.67	3.79	1.00	7.42

Fig. 12. The CC, RMSE and NRMSE of M3.

4.1. LSTM versus SPGP and RBF methods

Comparisons among CC, RMSE and NRMSE indicate that LSTM, RBF and SPGP can achieve continuous estimation. Based on the comprehensive analysis of the three evaluation critera, LSTM has the best performance among the three methods. However, SPGP or RBF has the better performance than LSTM in some joint, for example, in the estimation of *No*. 20 joint of M1 (see Fig. 6), the RMSE and NRMSE of SPGP (RMSE = 6.494°, NRMSE = 21.98%) are both less than LSTM (RMSE = 6.516°, NRMSE = 23.68%). For each movement, there are several results of SPGP or RBF better than LSTM (see Figs. 9, 10, 11, 12, 13 in Appendix). But the statistical analysis of CC, RMSE as well as NRMSE (see Figs. 7, 8, 9) show that LSTM can perform better in continuous estimation of 20 finger joint angles than SPGP

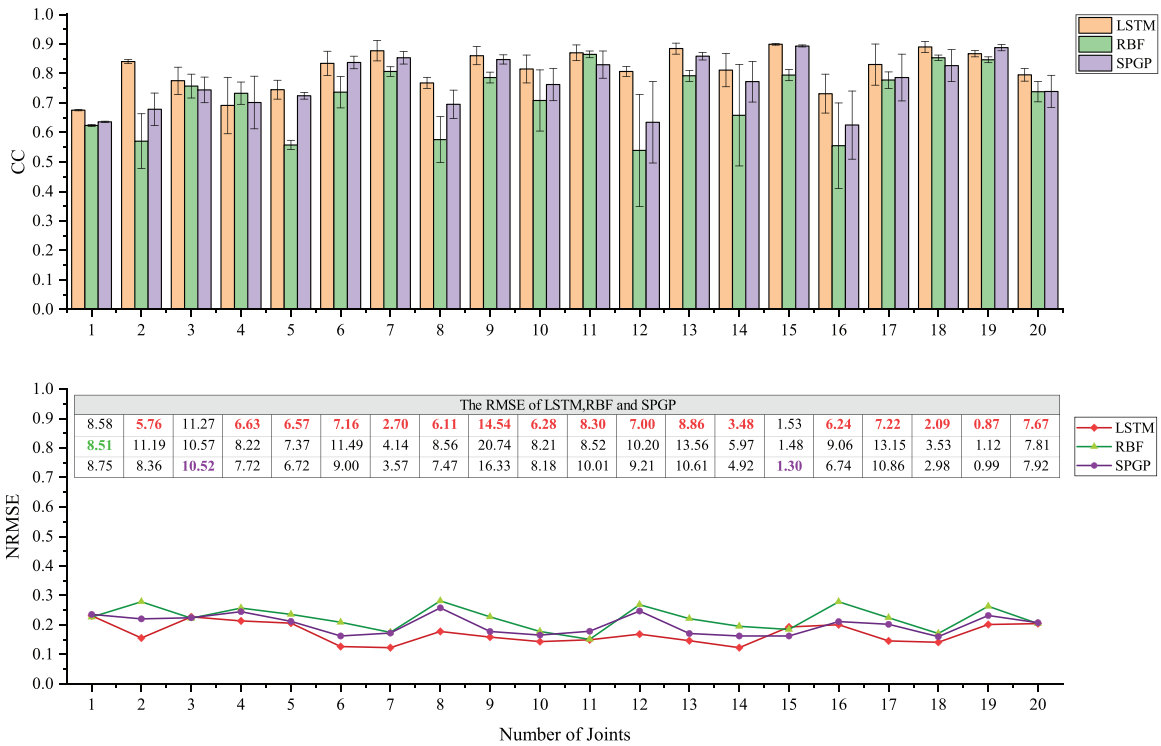


Fig. 13. The CC, RMSE and NRMSE of M4.

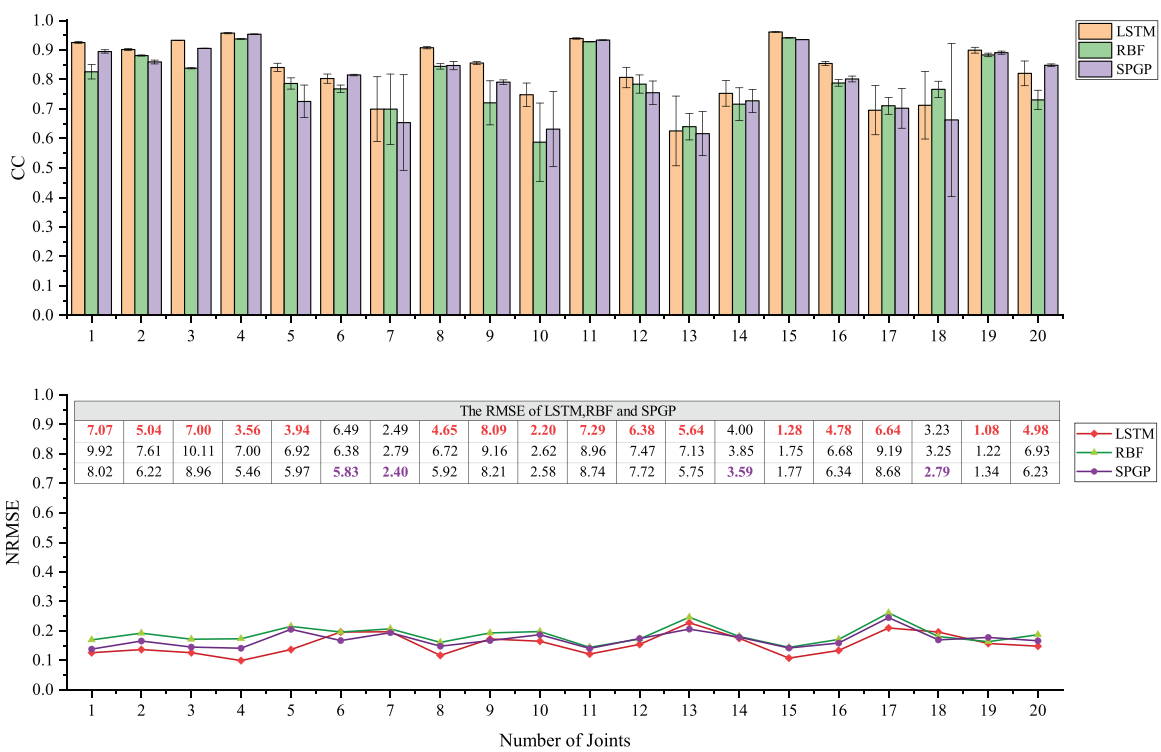
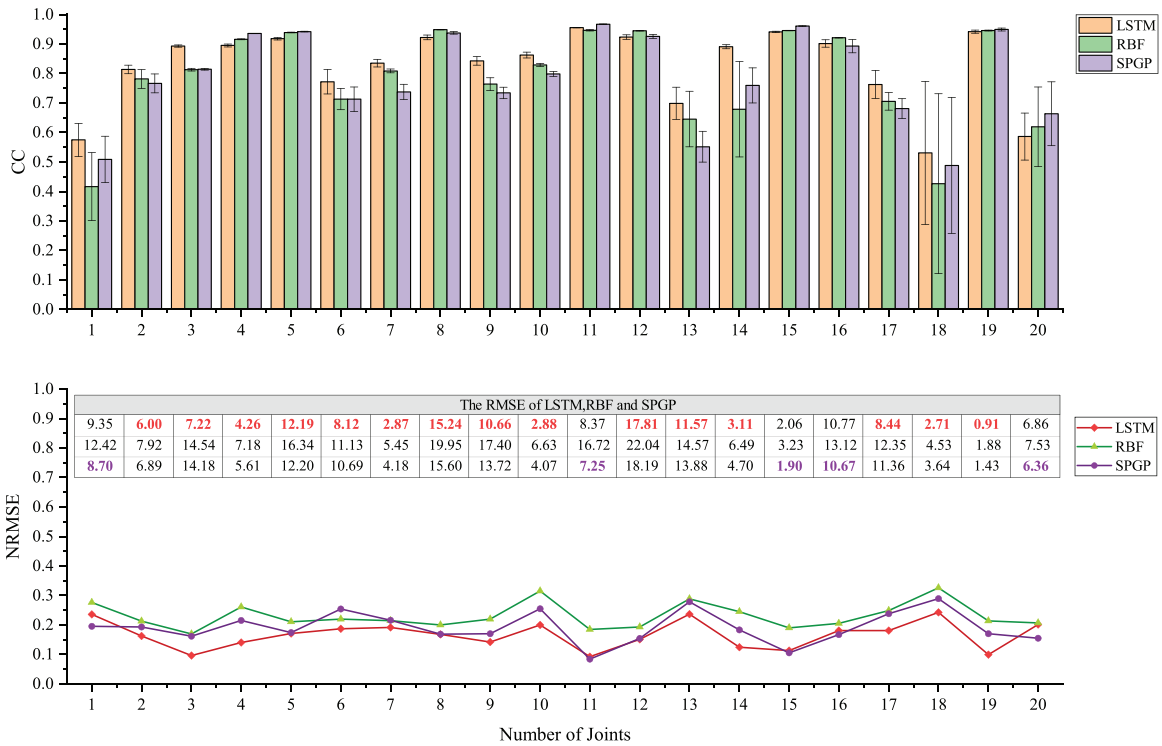


Fig. 14. The CC, RMSE and NRMSE of M5.

and RBF. In fact, during training, LSTM and RBF also need amount time, but when the model training is completed, motion information can be quickly extracted from sEMG. Considering in practical application, LSTM is more suitable in continuous estimation and myoelectric control.

4.2. Grasp movements and subjects

This study chose 6 grasp movements with objects of different diameters. The type and intensity of the grasp will affect the estimation results. It is deduced from the results that the continuous estimation for large diameter grasp (M1) or power grasp (M2) is



The RMSE of LSTM,RBF and SPGP

9.35	6.00	7.22	4.26	12.19	8.12	2.87	15.24	10.66	2.88	8.37	17.81	11.57	3.11	2.06	10.77	8.44	2.71	0.91	6.86
12.42	7.92	14.54	7.18	16.34	11.13	5.45	19.95	17.40	6.63	16.72	22.04	14.57	6.49	3.23	13.12	12.35	4.53	1.88	7.53
8.70	6.89	14.18	5.61	12.20	10.69	4.18	15.60	13.72	4.07	7.25	18.19	13.88	4.70	1.90	10.67	11.36	3.64	1.43	6.36

Fig. 15. The CC, RMSE and NRMSE of M6.

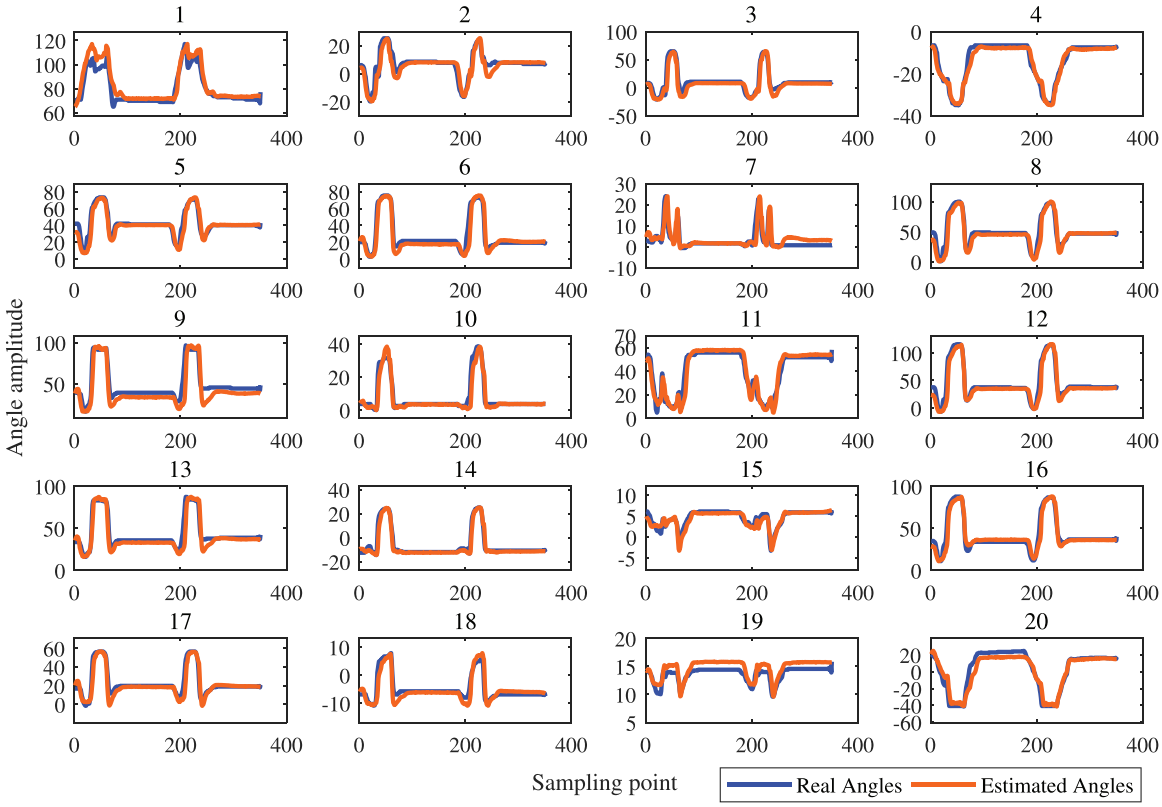


Fig. 16. Estimated and real joint angles for M2 of subject 1.

better than stick grasp (M4) or disk grasp (M5). When the subjects grasp with force, the sEMG signals generated by the muscle is more obvious, and the reflected motion characteristics are more obvious, so the estimation results of these movements are better, referred to Fig. 11 in the Appendix. But no matter which type of grasp, there are bad results in some joints because of the different importance of different joints in the grasp movements. For the grasp of the cylinder, the metacarpal joints are of the greatest importance, followed by the proximal joints or the distal joints. For example, as shown in Fig. 6, the results of *No.* 6 (index proximal joint) or *No.* 7 (index

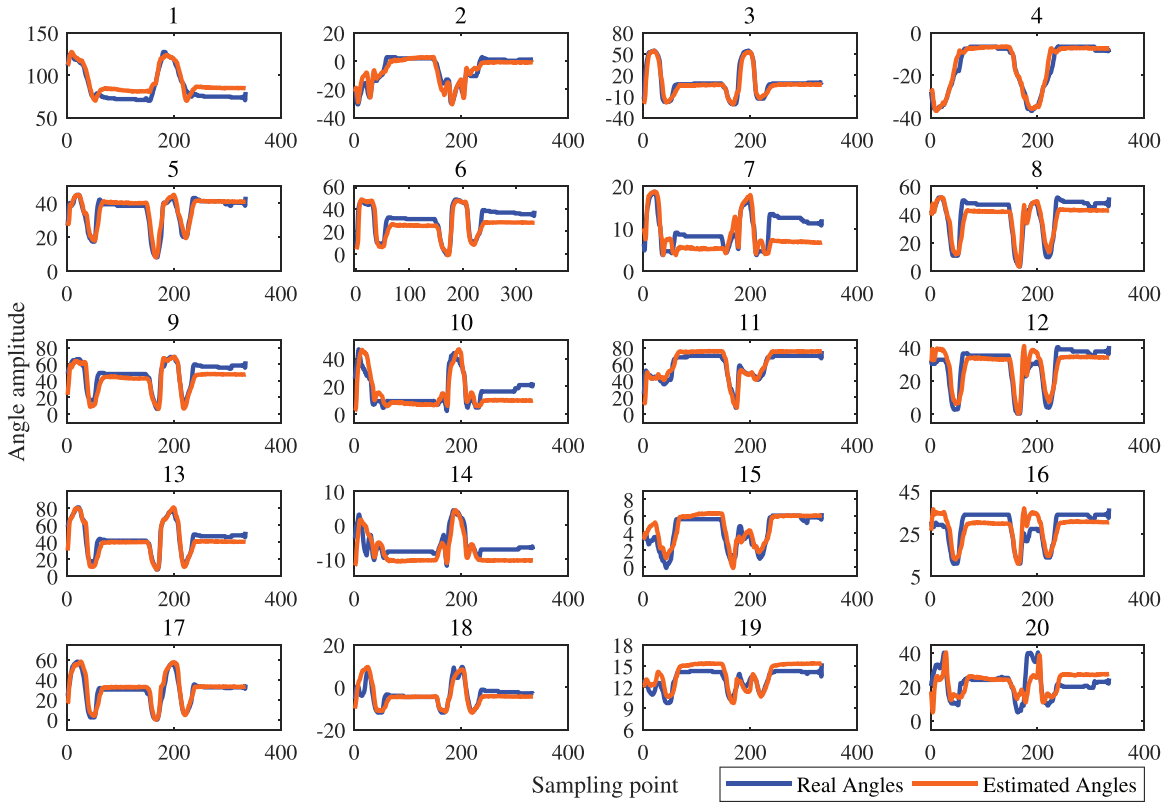


Fig. 17. Estimated and real joint angles for M3 of subject 1.

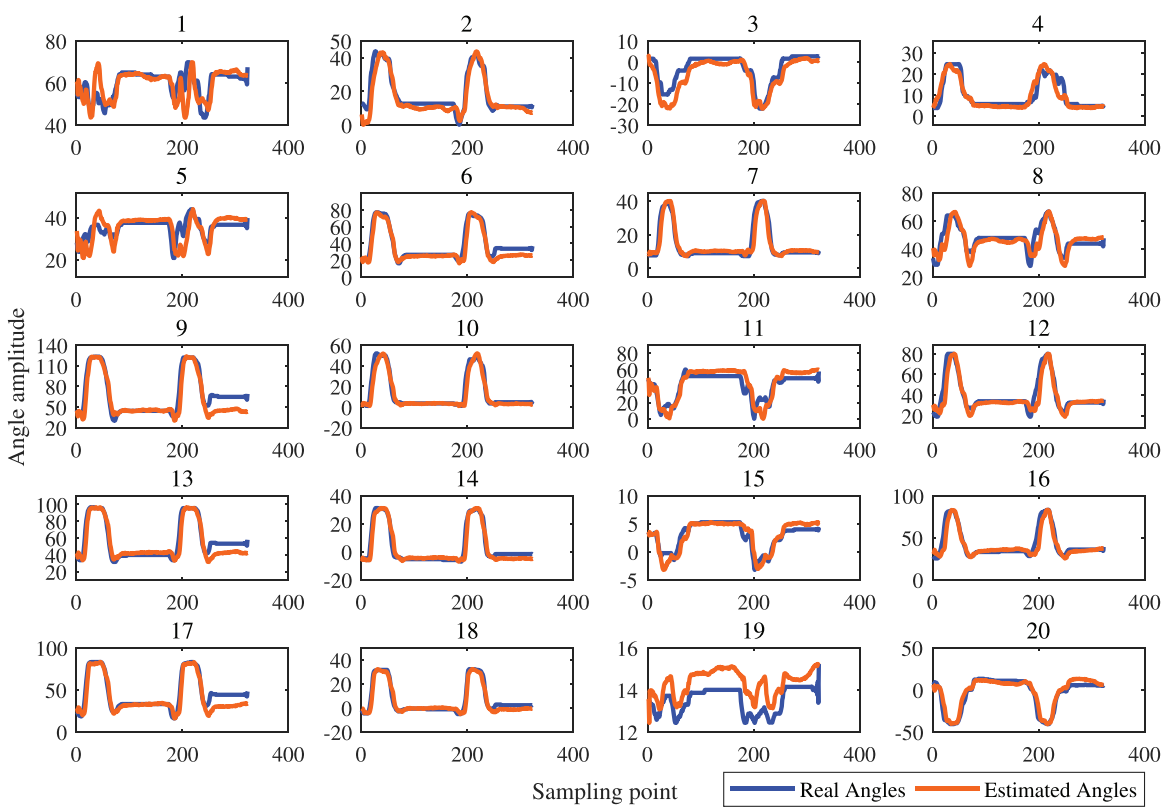


Fig. 18. Estimated and real joint angles for M4 of subject 1.

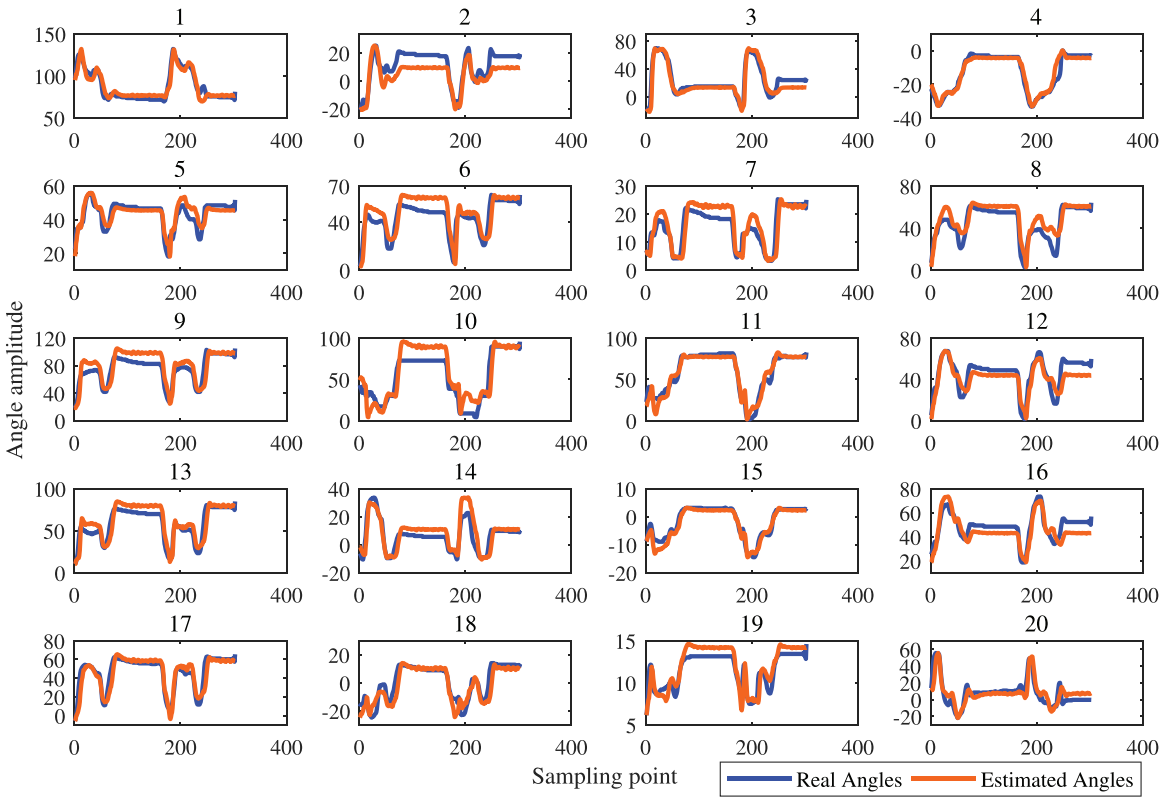


Fig. 19. Estimated and real joint angles for M5 of subject 1.

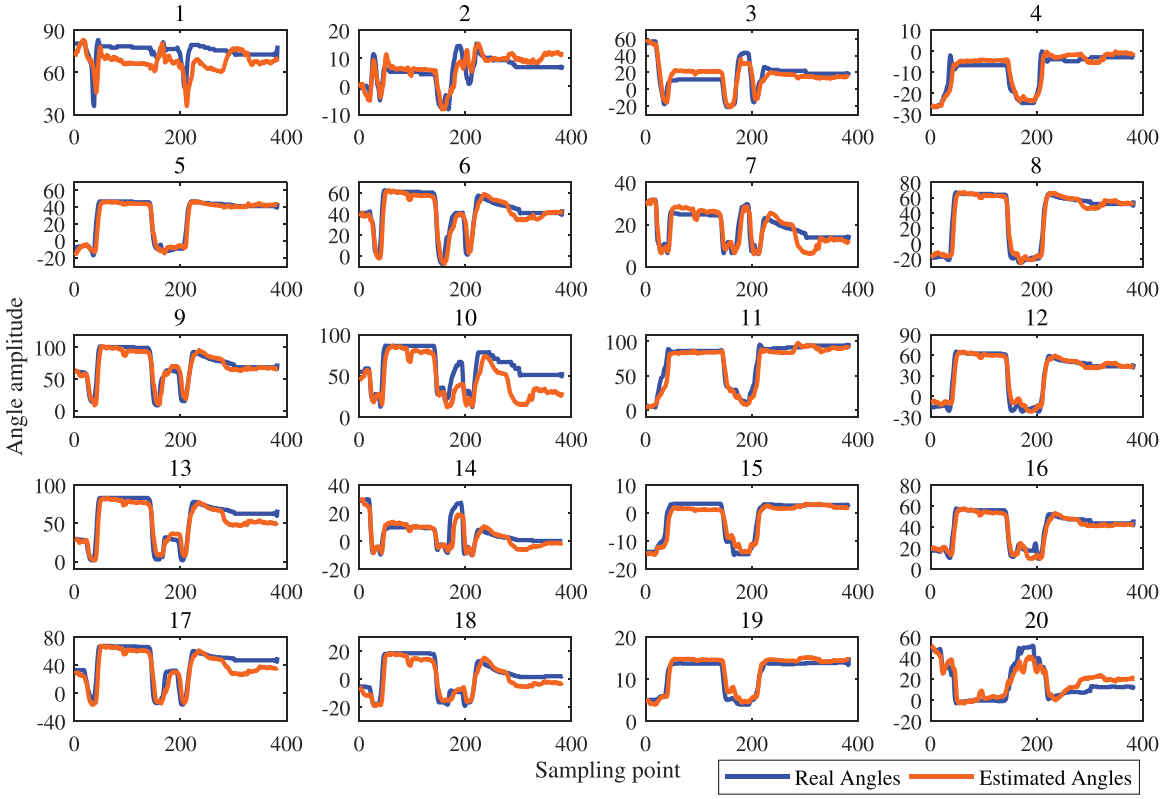


Fig. 20. Estimated and measured joint angles for M6 of subject 1.

distal joint) are worse than *No*. 5 (index distal joint). Nevertheless, in the estimation of these joint angles, the performance of LSTM is more prominent than the other two methods. Besides, the quality of the data acquired will also affect the estimated results. For example, the estimation of *No*. 10 or *No*. 14 joint of M1 shown in Fig. 10 is poor because of the original collected signals confusion. Apart from this, we can observe from the maps of real angles and estimated angles that the results of M4–M6 are worse than M1–M3. The possible reason may be data deteriorates as the subjects complete more movements.

5. Conclusion

We compared multiple methods for continuous estimation of grasp movement, and found that LSTM held the best performance. We not only realize the continuous estimation of finger joint angles but also increase the number of finger joint angles from about 10 to 20.

This study may promote the research development on synchronous and proportional myoelectric control. We find the bright prospects of LSTM which can be used in bioelectrical signals processing and human-machine-interaction. But considering the different applications, the method should be personalized and optimized.

In the future, we will explore other methods compared with LSTM in order to get a better one and continue to work on the estimation of multi-degree of freedom synchronously.

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Declaration of Competing Interest

None declared.

Appendix A.

In order to intuitively and comprehensively demonstrate the performance of LSTM, RBF and SPGP in the continuous estimation, the results with CC, RMSE and NRMSE of M2–M6 are shown in Figs. 11, 12, 13, 14, 15 . According to the comparison of these results, we can conclude that LSTM performs better than RBF and SPGP in grasp movements estimation. And the performance of LSTM for M2–M6 of subject 1 are shown in Figs. 16, 17, 18, 19, 20 . All the units of “Angles ampulitude” in figures and the units of “RMSE” in the tables are degree measure, “◦”.

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